

Basic Concepts: Transactional Database

❑ Transactional Database (TDB)

- ❑ Each transaction is associated with an identifier, called a TID.
- ❑ May also have counts associated with each item sold

TID	Items bought
1	Beer, Nuts, Diaper
2	Beer, Coffee, Diaper
3	Beer, Diaper, Eggs
4	Nuts, Eggs, Milk
5	Nuts, Coffee, Diaper, Eggs, Milk

Basic Concepts: k-Itemsets and Their Supports

- **Itemset**: A set of one or more items

$$I = \{I_1, I_2, \dots, I_m\}$$

- **k-itemset**: An itemset containing k items:

$$X = \{x_1, \dots, x_k\}$$

- Ex. {Beer, Nuts, Diaper} is a 3-itemset

- **Absolute support (count)**

- $\text{sup}\{X\}$ = occurrences of an itemset X

- Ex. $\text{sup}\{\text{Beer}\} = 3$
- Ex. $\text{sup}\{\text{Diaper}\} = 4$
- Ex. $\text{sup}\{\text{Beer}, \text{Diaper}\} = 3$
- Ex. $\text{sup}\{\text{Beer}, \text{Eggs}\} = 1$

TID	Items bought
1	Beer, Nuts, Diaper
2	Beer, Coffee, Diaper
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- **Relative support**

- $s\{X\}$ = The fraction of transactions that contains X (i.e., the **probability** that a transaction contains X)
- Ex. $s\{\text{Beer}\} = 3/5 = 60\%$
- Ex. $s\{\text{Diaper}\} = 4/5 = 80\%$
- Ex. $s\{\text{Beer}, \text{Eggs}\} = 1/5 = 20\%$